

Histology

Nervous tissue

Nervous tissue consists of:

- Central nervous system (CNS) including brain and spinal cord.
- Peripheral nervous system (PNS) including nerve axons, nerve endings and ganglia.

Nerve cell or Neuron

A neuron is the structural and functioning unit of the nervous system; specialized to receive, integrate, and transmit information. They are in G0 of cell cycle.

Neurons are composed of: cell body (perikaryon) and processes.

Histology of the nerve cell

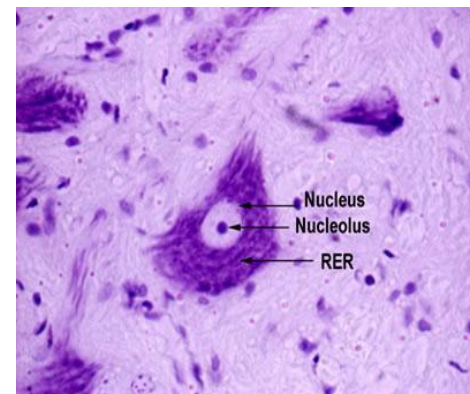
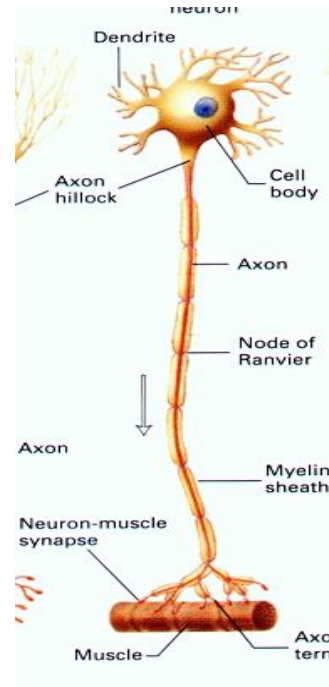
LM

Large cell body.

Large, central pale open face nucleus with prominent nucleolus.

Basophilic cytoplasm rich in Nissl granules in the body and dendrites, but not in the axon.

Inclusions: lipofuscin pigment and melanin pigment. The lipofuscin pigment represents “wear and tear” product that cannot be eliminated by lysosomal digestion. Melanin pigments are present in few parts of the C.N.S. e.g. substantia nigra.



EM

- The nuclear chromatin is extended “euchromatic”.
- Nissl bodies appear as flattened cisterns of rER with numerous free and attached ribosomes.
- Neurofilaments and neurotubules form the cytoskeleton of the cell.
- Golgi complex and abundant mitochondria.
- Primary lysosomes are common, and secondary lysosomes increase in number with age.

Processes of the nerve cell:

Neuro 25-26

1. **Axon** is single, thin, and long process.

Its function is to transmit nerve impulses. The axoplasm containing microtubules, neurofilaments, numerous mitochondria and axoplasmic vesicles. It lacks Golgi, RER, and ribosome

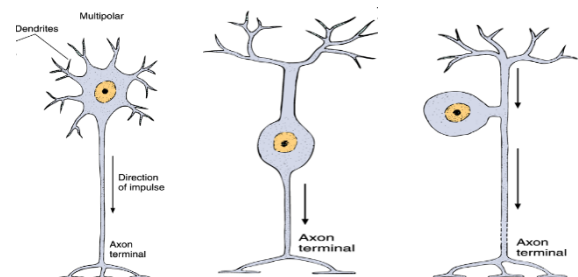
2. **Dendrites** are multiple thick and short cytoplasmic extensions that extend in any direction and branch repeatedly (with the final branches being smaller) to increase the surface area.

Dendrites	Axon
Multiple	Single
Receive impulse	Transmit impulse
Thick and short	Long (Golgi type 1) Short (Golgi type 2)
Lack only Golgi	The axoplasm contains microtubules, neurofilaments, numerous mitochondria and axoplasmic neurotransmitter vesicles. Lack ribosomes, Golgi

Both axon and dendrites have mitochondria, microtubules and neurofilaments.

Types of nerve cell according to the number processes number:

1. **Multipolar**: the most common type, has number of dendrites and only one axon. Usually, the axon is long, and the neuron is called Golgi type 1 while some neurons have short axons called Golgi type 2. e.g anterior horn cells (AHCs) and autonomic ganglia.
2. **Bipolar**: usually oval with two processes, one of them acts as a dendrite and the other acts as an axon. It is present in the special sense organs e.g retina.
3. **PseudoUnipolar**: only one process arises from the cell body and bifurcates not far from it into two processes, one acts as an axon and the other as a dendrite. They are large size neurons present in the spinal ganglia



Types of Nerve cell according to function:

- 1- Sensory
- 2- Motor
- 3- Interneuron : neurons modifies responses of other neurons

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The cell body of the neuron is present within the central nervous system protected in the brain and spinal cord. As it is considered as **permanent tissue**. The peripheral nerves are axons of these cells.

Types of nerves

1. Autonomic nerves:

These nerves control the involuntary or partially voluntary activities of your body.

2. Motor nerves:

These nerves control movements and actions by passing information from brain and spinal cord to your muscles.

3. Sensory nerves:

These nerves relay information from skin and muscles back to spinal cord and brain. The information is then processed to get pain and other sensations.

Neuroglia

In the PNS, there is beside nerve cells, ordinary connective tissue fibers and cells are present. In the CNS, there is no connective tissue, but instead there is a special type of binding cells “neuroglia cells”, which assist the nerve cells in performing their functions.

Types:

1- **Oligodendrocytes** for myelination in CNS.

2- **Microglia** act as macrophage

3- **Astrocytes** both fibrous and protoplasmic forming the blood brain barrier and share in impulse transmission

4- **Ependymal** cells lining of fluid-filled cavities, formation of CSF.

Other cells in the PNS:

1- **Schwann cells** for myelination and regeneration in PNS.

2- **Satellite cells** surround neurons in sensory ganglia to provide support and nutrients.

Peripheral nervous system

Synapses

The Synapse is the point of connection between two neurons or between a neuron and a muscle or gland or a highly specialized contact point which exists at an

axon terminal, allowing the signal to pass from cell to cell. Electrochemical communication between neurons takes place at these junctions.

Structure of synapse:

A. Presynaptic membrane:

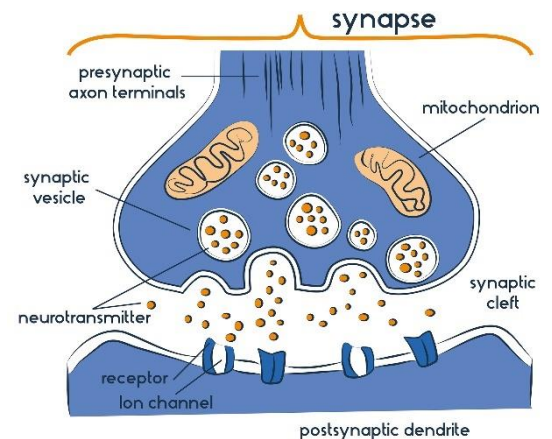
Formed of terminal button found at the end of the axon

B. Postsynaptic membrane:

Composed of a segment of dendrite or cell body.

C. Synaptic cleft:

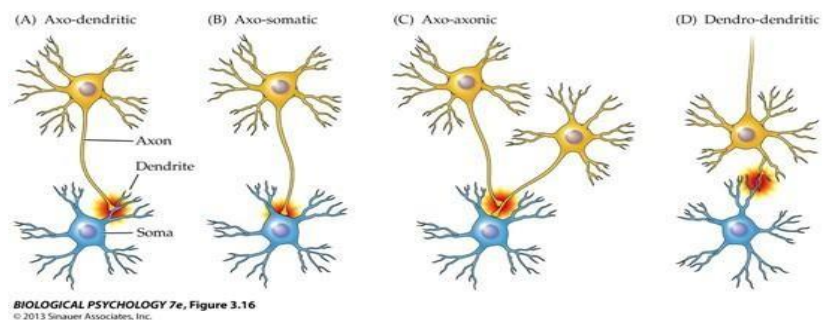
The space between presynaptic and postsynaptic membranes.



Types of synapses

- Different types of synapses between two neurons:

- 1- Axo-somatic
- 2- Axo-dendritic
- 3- Axo-axonic
- 4- Dendro-dendritic.



- According to the transfer of information:

1. Chemical synapses: are common. The nerve impulse is transmitted chemically via neurotransmitters.
2. Electrical synapse: rare. The nerve impulse is transmitted electrically via channel proteins and has gap junctions through which ions can pass freely, so conducting the nerve impulse directly and much rapidly.
3. Mixed synapse: using both chemical and electrical methods.

Ganglia

Definition: it is an aggregation of nerve cell bodies outside the C.N.S

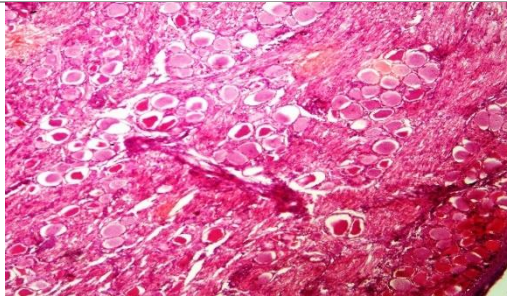
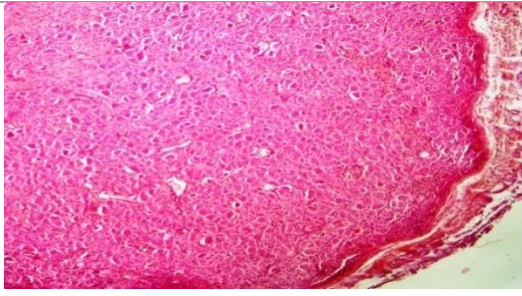
Types:

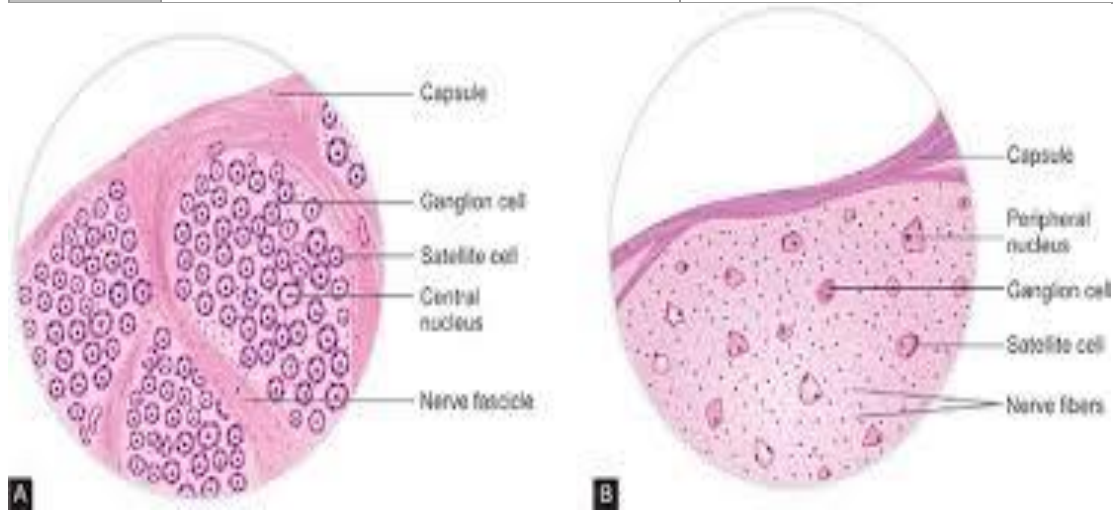
Spinal ganglia (dorsal root ganglia)

Autonomic (sympathetic-parasympathetic)

	Spinal ganglia	Autonomic ganglia
Site	Dorsal root ganglia	Sympathetic and parasympathetic

Neuro 25-26

Nerve cells	Large, pseudounipolar, cell bodies lie peripheral in groups separated by bundles of nerve fibers	Small, multipolar, the nerve cells and nerve fibers are intermingled and not present in groups
Nerve fibers	Central, myelinated, or unmyelinated	Non-myelinated
Satellite cells	Present	Fewer
Synapse	Rare	Numerous
		



Peripheral nerve fibers

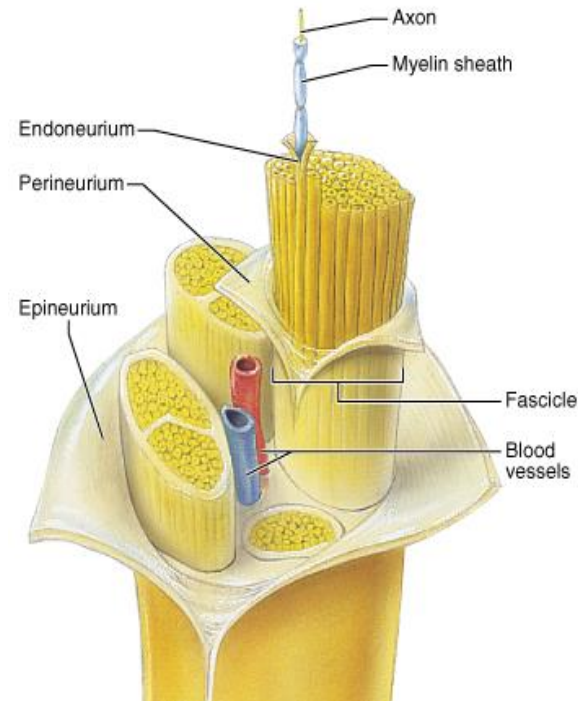
The axon usually arises from the soma at an area devoid of Nissl granules called an axon hillock. Its plasma membrane is called the axolemma, and the cytoplasm is called axoplasm. The axoplasm does not contain the Nissl or Golgi apparatus, but it contains mitochondria, microtubules, and neurofilaments. Axons are either myelinated or unmyelinated.

Although connective tissue in the CNS is nearly limited to the meninges, it is a prominent feature in the PNS. A nerve is not a single axon but rather is bundles of axons (nerve fibers).

Each axon is surrounded by a CT sheath called endoneurium.

Each bundle is surrounded by a CT sheath or perineurium.

All of these bundles together are surrounded by CT epineurium rich with collagen fibers.



In the osmium-stained section of a peripheral nerve, myelin appears as black circles.

